Online Resource 1

Supplementary material for the article "AXIOM: A Shared Graph-Attention Controller for Scalable Multi-Agent 5G Slice Admission"

Wireless Personal Communications

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S1. Overview

This supplement gives the detail behind the main paper's verification runs and additional results for AXIOM, the Adaptive Cross-cell Integrity-aware Open Radio Access Network Multi-agent slice-admission controller. Every pass criterion was fixed in a pre-registration committed before the run it judged. The baselines are a single-agent deep Q-network (DQN-SA), independent per-gNB deep Q-networks (DQN-I), and multi-agent proximal policy optimisation (MAPPO), where DQN denotes a deep Q-network in general. AXIOM's encoder is a graph attention network (GAT), and its variant with a per-node multi-layer perceptron (MLP) encoder is called the MLP variant. Table S1 lists the other acronyms used in this supplement.

Table S1 Acronyms used in this supplement

Acronym	Meaning	Acronym	Meaning
CI	Confidence interval	PRB	Physical resource block
CPU	Central processing unit	RAN	Radio Access Network
DP	Differential privacy	RF	Radio frequency
DP-SGD	Differentially private stochastic gradient descent	RIC	RAN Intelligent Controller
eMBB	Enhanced Mobile Broadband	SDR	Software-defined radio
FedAvg	Federated averaging	SLA	Service Level Agreement
gNB	gNodeB, the 5G base station	TDD	Time-division duplex
GPU	Graphics processing unit	UDP	User Datagram Protocol
mMTC	Massive Machine-Type Communication	UE	User equipment
OAI	OpenAirInterface	URLLC	Ultra-Reliable Low-Latency Communication
O-RAN	Open Radio Access Network		

S2. Verification runs

We reran eleven reported settings on their original seeds in fresh processes, and each reproduced its stored per-seed results exactly, as did the Env A oracle rule on all 30 seeds. Sixteen runs that no earlier rerun covered matched their stored per-seed results exactly, in Env A at seven, nineteen, and thirty-seven gNBs and in Env B at seven gNBs. DQN-SA is the one exception, because its training code builds the network before it sets the seed. Dropping the affected seeds changes no comparison's direction. With its initialisation seeded, DQN-SA then reproduced exactly across processes.

Four further runs retrained every policy on seeds from 3000 onwards, never used before, and 81 of their 90 checks stated in advance held. In Env A AXIOM reached 98.2% of the oracle reward at three gNBs and at nineteen gNBs kept 95.6% priority-slice compliance against 47.8% to 65.5% for the baselines. At seven gNBs on the hex layout and at nineteen on the fully-connected layout AXIOM kept 99.0% and 95.7% priority-slice compliance and beat every baseline. Federation again showed no significant cost, at +0.077 with 95% CI [−1.03, 1.27]. Fresh seeds therefore confirmed the main findings of the paper. At

seven and nineteen gNBs in Env A AXIOM kept 93.4% to 100% priority-slice compliance and beat every baseline under every layout.

Five of the nine failed checks concern the baselines rather than AXIOM. On fresh seeds at seven gNBs DQN-I and DQN-SA kept up to 83.3% priority-slice compliance, above the 80% bound we had set. The other four concern small margins of AXIOM itself. Under the fully-connected layout at seven gNBs two weak AXIOM seeds left its leads short of significance after Holm correction, and a twenty-seed rerun of that layout then held all six tests at Holm-adjusted $p \leq 0.007$. In one ten-seed load sample two weak AXIOM seeds also left it 3.2% short of the oracle, a gap that a twenty-seed rerun did not reproduce. AXIOM's small lead over DQN-SA in Env A at three gNBs did not recur, at margins of -0.021 and 0.159 reward per step on 20 and 40 new seeds.

An independent script with its own log reader and Holm correction recomputes every results table and key sentence of both documents from the raw logs, and all of its checks pass. Fig. S1 sets each reported comparison with a learned baseline beside its fresh-seed counterpart. Of the 38 reported differences whose 95% CI excludes zero, 35 keep a CI on the same side on fresh seeds. The three exceptions all concern cluster-level compliance in Env B. AXIOM's deficits of 9.2 points to DQN-SA and 4.8 points to DQN-I at seven gNBs become 4.2 points and a 0.8-point lead. Its lead over MAPPO at nineteen gNBs falls to 2.9 points, and all three fresh CIs span zero.

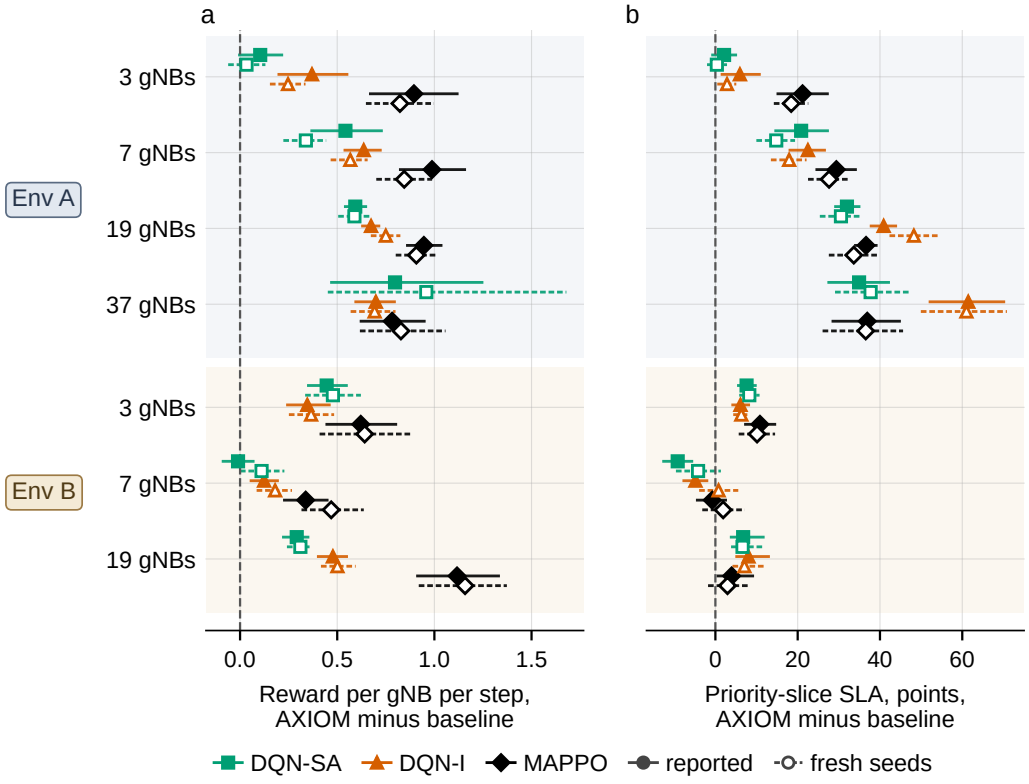


Fig. S1 AXIOM minus each learned baseline, with 95% CIs. Solid lines and filled circles are the reported runs, dashed lines and hollow diamonds the fresh seeds. a, Reward per gNB. b, Cluster-level priority-slice compliance

S3. Live testbed and deployment check

Every live measurement comes from the single-host, software-radio O-RAN stack of Fig. S2. It runs without a near-real-time RAN Intelligent Controller (RIC), because the ORANSlice fork applies slicing control directly through a lightweight User Datagram Protocol (UDP) E2 agent. AXIOM itself is not deployed on the live testbed. Its multi-gNB policy needs at least two live gNBs, and our two-gNB attempt registered both with the core but did not sustain traffic. The testbed therefore validates the pipeline AXIOM would use, namely the control path over the O-RAN E2 interface, PRB-ratio actuation, and real-time inference. It does so with a single-agent DQN checkpoint trained offline for this rig, at one and at two user equipment (UE) devices per slice. A capacity probe showed every slice served at essentially

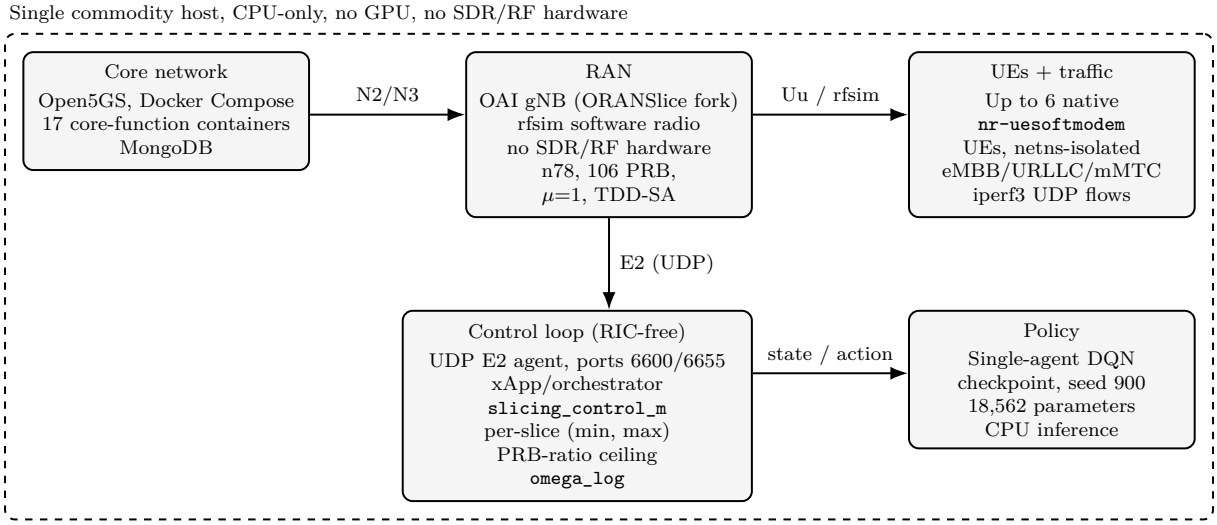


Fig. S2 Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from software-defined-radio hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig

100% of offered demand with 0.000% loss. The checkpoint's real accept and reject decisions, typically blocking mMTC alone, ran correctly through the E2 loop. Because the rig never ran short of capacity, this shows that the pipeline works, not admission under contention. A scheduler-configuration error that granted fewer PRBs than the gNB scheduler's five-PRB floor was fixed at the root. Instrumentation then confirmed zero below-floor grants across more than 100,000 readings.

S4. Setup, safeguards, and analysis plans

Algorithm 1 of the main paper states AXIOM's training loop, and its parameter table lists the simulation and training settings.

A sensitivity study repeats the learned policies and the oracle rule in Env A on 10 seeds at 102% and 107% of the nominal load and with changed slice weights, eMBB lowered from 3.5 to 2.5 and mMTC raised from 0.3 to 0.6. A second repeats AXIOM in Env B on 10 seeds at learning rates of 3×10^{-4} and 3×10^{-3} and with an encoder twice as wide. The federated campaign runs in Env A on seeds 900 to 909, with noise multipliers $\sigma \in \{0, 0.5, 1, 2, 4\}$ and a replication on seeds 1000 to 1009. Each round runs 50 local steps of federated averaging (FedAvg) with a gradient-clip norm of 1.0.

A validation screen checks every trained checkpoint of every learned policy before evaluation. Each plays five greedy episodes on its own training stream and fails if it blocks no request or if fewer than 90% of its blocks target mMTC. Every seed is reported whether or not it passes. In Env B the reward-optimal policy also sheds eMBB, so an amendment recorded before any result was read makes the no-blocking criterion the primary screen there. The two Env B constants were calibrated on pilot seeds with rules only. The main pre-registration fixed them together with AXIOM's learning rate and skip connection, all chosen on pilot seeds 2000 to 2009.

AXIOM's encoder is a graph-attention network over the gNB graph. At every control step each gNB forms its own feature row $\mathbf{x}_i$ of PRB-used ratio, congestion level, and normalised queue length. A two-layer network with H=4 heads then produces one embedding per gNB. Each head h uses a learned source vector $\mathbf{a}_s^h$ and destination vector $\mathbf{a}_d^h$ on the shared projection $\mathbf{W}^h$. Here LeakyReLU, the leaky rectified linear unit, has slope 0.2 for negative inputs, and $\mathcal{N}_i$ is the set of neighbours of node i,

$$\varepsilon_{ij}^h = \mathrm{LeakyReLU}\big(\mathbf{a}_s^{h\top}\mathbf{W}^h\mathbf{x}_i + \mathbf{a}_d^{h\top}\mathbf{W}^h\mathbf{x}_j\big), \qquad \alpha_{ij}^h = \frac{\exp\varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i}\exp\varepsilon_{il}^h}. \tag{1}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h is its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an exponential linear unit (ELU), while the final layer averages them. A per-layer normalisation $\mathrm{LN}(\cdot)$ then rescales each node's embedding to zero mean and unit variance, giving the per-gNB embedding $\mathbf{z}_i$,

$$\mathbf{z}_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^h\,\mathbf{W}^h\mathbf{x}_j\Big). \tag{2}$$

Every agent's loss updates this shared encoder, so training sees the full cluster state. At execution each agent reads only its own row $\mathbf{z}_i$ to decide its pending requests. An earlier variant mixed each gNB's summed per-request values with a QMIX-style monotonic mixer. Its target changed scale with the request count, and its mean training loss diverged from about 76 to about 409. Treating each request as its own sample removed that dependence.

In the federated variant each client's encoder still reads the full joint node-feature snapshot. Differential privacy bounds how much any one client's training data can change what the shared model reveals. Each client applies a differentially private stochastic gradient descent (DP-SGD) step, clipping its minibatch gradient to norm C_{dp} and adding Gaussian noise of standard deviation σC_{dp}. Zero-concentrated differential privacy (zCDP) composes this over the m noisy steps a client takes, with ε and δ the usual privacy-loss and failure-probability parameters,

$$\rho = \frac{2m}{\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{3}$$

We credit no amplification by subsampling, because minibatches come from a replay buffer that reuses records, so the bound is conservative. The noise protects each client's admission decisions and rewards, but not the congestion features that every encoder reads by design.

The trivial-policy check evaluates the simplest rules before trusting any learned policy. In our first simulator every slice was served independently against its own ceiling. There, accepting every request earned 14.45 reward per step and kept URLLC and eMBB at 100% SLA compliance, more than any learned policy, so admission control had nothing to do. The shared-budget simulator of the main paper removes that loophole. Over the same 30 held-out seeds, accepting every request now earns 9.89 reward per step and lets eMBB compliance fall to 56.8%, whereas shedding every mMTC request earns 12.84 and keeps URLLC and eMBB at 100%. Replication on a disjoint seed sample caught one artefact. Queue backlog carried over between episodes, so one early transient spoiled every later episode for every policy, the oracle rule included. Every result in this paper comes from the corrected simulator, which empties all queues at the start of each episode.

Each analysis plan was committed to the project's version-control history before the runs it governs, with one exception. The main campaign reuses Env A results of DQN-SA, DQN-I, and the oracle rule from an earlier rerun, M63. They cover three gNBs on the primary and replication seeds, the sensitivity study, seven gNBs, and 15 of the 30 seeds at nineteen gNBs. The M63 plan was committed one minute after that rerun was launched and before any of its results were read. Two development stages are also disclosed, although neither changed a reported result. A protocol, M67, tested checkpoint selection and both encoders on new seeds, found no decisive gain over DQN-SA, and left AXIOM unchanged. Exploratory probes on seeds 900 to 904 and pilot seeds 2000 to 2004 suggested the load-shift effect of Section S7 and the hindsight bound of the main paper's Env B results before their reported runs. Three figures describe stored runs after the results were known, with no plan or test of their own. The main paper maps per-gNB compliance at nineteen gNBs. Section S2 sets reported beside fresh-seed differences, and Section S7 shows AXIOM's learned eMBB shedding after replaying every stored evaluation exactly.

An AI assistant, Anthropic's Opus agent, was used to help write the analysis and plotting scripts and to edit the manuscript text. The authors made all research decisions and take full responsibility for the content of this paper.

S5. Three gNBs, ablations, federation, and disruption

The MLP variant reads only a gNB's own features. In Env A it ties AXIOM, which earns 0.058 more reward per step at three gNBs with 95% CI [−0.17, 0.26] and 0.31 more at nineteen gNBs on the ring layout, with 7 of 12 seeds tied. In Env B it earns slightly more, 0.116 per step at three gNBs in 17 of 30 seeds at $p = 0.45$. At seven gNBs it earns 1.12 and 0.95 more, but neither lead survives Holm correction. On ten fresh seeds at seven gNBs in Env B, however, it led by 1.38 reward per step, in 9 of 10 seeds at $p = 0.006$.

A disjoint replication on seeds 1000 to 1009 is stronger for AXIOM. There it earns 1.083, 0.799, and 3.806 more reward per step than DQN-SA, DQN-I, and MAPPO. It sits only 0.018 below the oracle on those seeds. Pooled over all 40 seeds its margins over the three baselines are 0.504, 1.032, and 2.965 reward per step at Holm-adjusted $p \leq 0.002$. Fresh seeds in Section S2 do not reproduce the lead over DQN-SA, however. In Env B at three gNBs the disjoint ten-seed replication points the same way, with AXIOM winning 7 or 8 of 10 seeds against each, but no lead survives correction there.

Under 102% and 107% load and the changed slice weights AXIOM stays within 0.13 of the oracle rule's reward. Its leads over both DQN baselines are positive under every condition, but at ten seeds only two survive correction. They are its margin over DQN-I at 107% load, 1.073 reward per step at $p = 0.006$, and over DQN-SA under the changed weights, 1.264 at $p = 0.041$.

The comparison also holds when the baselines are made stronger. A grid of 24 MAPPO configurations, chosen on pilot seeds and allowing three times the training budget, kept the reported setting in Env B and selected a longer-trained one in Env A. There MAPPO rises from 10.013 to 11.448 reward per step at three gNBs and from 60.6% to 69.6% priority-slice compliance at nineteen gNBs on the ring layout. AXIOM still beats it in every setting tested, by 1.25 to 16.18 reward per step at Holm-adjusted $p < 0.001$, and by 2.42 in Env B at seven gNBs. With its initialisation properly seeded, DQN-SA keeps 80.3% and 63.9% priority-slice compliance at seven and nineteen gNBs, and AXIOM beats it under every layout at Holm-adjusted $p \leq 0.035$.

AXIOM's learning rate matters most as the cluster grows. At the original rate of 10^{-4} AXIOM earns only 6.496 per step in Env B on seeds 900 to 909, and $3{\times}10^{-4}$ reaches 7.220. The DQN baselines also lose reward when rerun at 10^{-4}, falling to 11.477 [10.61, 12.09] and 11.119 [10.73, 11.50] reward per step in Env A at three gNBs, with 95% CIs. A rate of $3{\times}10^{-3}$ and an encoder twice as wide both stay within 0.19 of the current AXIOM. At nineteen gNBs in Env A the original AXIOM kept priority-slice compliance at only 79.1%. Over the Env B seeds the original AXIOM earns 2.37 less per step than the current design in all ten seeds. The joint-target variant of AXIOM, trained beside it on fresh seeds, never differed significantly from it after Holm correction over eight tests. It reaches 12.831 reward per step in Env A at three gNBs and 99.96% priority-slice compliance at nineteen. On 60 further fresh seeds across four settings it was again never significantly different from AXIOM, ahead in three settings and behind at seven gNBs in Env B, so we keep the reported design.

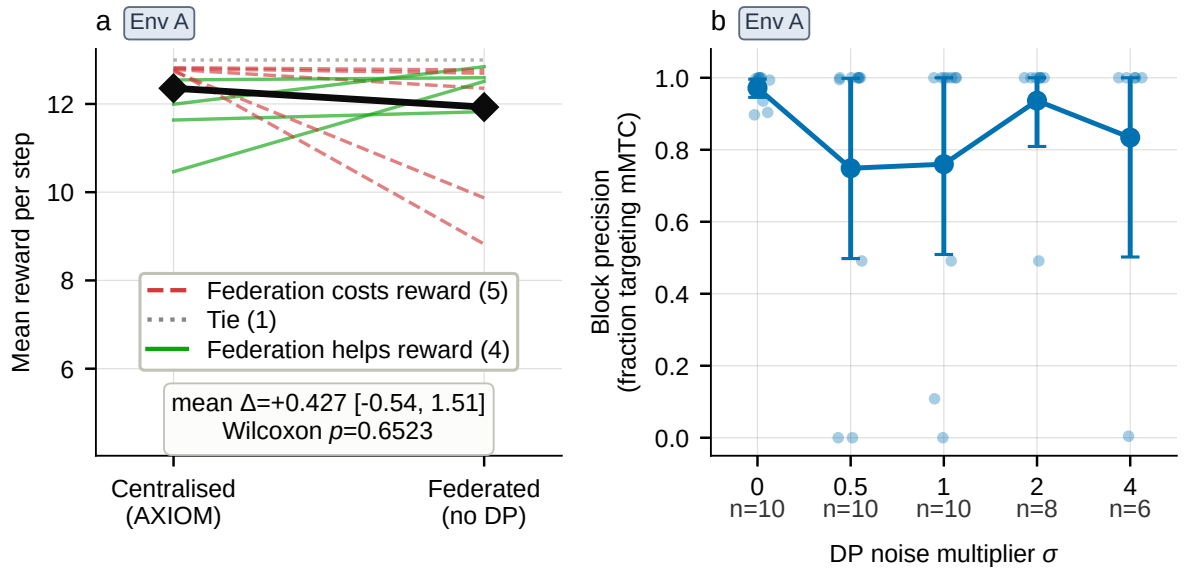


Fig. S3 Federated campaign in Env A, 10 seeds. a, Paired per-seed federation cost on mean reward per step, not significant at $p = 0.65$. b, Block precision versus DP noise multiplier σ, over the seeds that block

Fig. S3 shows the federated campaign in Env A on ten seeds. With federation and no noise, mean reward per step on the ten shared seeds is 12.356 for centralised AXIOM and 11.929 for the federated variant. That is a difference of +0.427 with 95% CI [−0.54, 1.51] and $p = 0.65$, so federation shows no significant cost on these seeds. The disjoint replication shows a larger cost of +0.968, with 95% CI [0.18, 1.96], which falls just short of significance at $p = 0.055$. FedProx, which adds a proximal term of weight μ to each client's loss, gives no consistent picture on three seeds, with μ=0.01 above FedAvg and μ of 0.1 and 1.0 below it.

DP-SGD noise on every step lowers reward by 0.77 to 1.25 per step at every tested noise multiplier. It also lowers eMBB compliance to between 74.3% and 83.0%, and two and four of the ten seeds collapse at σ of 2 and 4. None of these drops survives Holm correction at ten seeds. Each client takes 17,969 noisy steps over its training run. At δ=10^{-5} Eq. (3) then gives ε of about 146,000, 37,000, 9,600, and

2,600 for σ of 0.5, 1, 2, and 4. Per-step DP-SGD therefore gives no meaningful guarantee here. A useful one would need noise on the roughly 360 per-round model updates instead.

Each policy's frozen Env A checkpoint also faced three disruptions at evaluation time. A gNB dropout forces one gNB to reject everything and blanks its feature row. A demand spike multiplies the arrival rate, and an agent churn swaps one gNB's decisions for an untrained copy of its policy. Dropout and churn windows last 10%, 30%, or 60% of the episode. Spikes multiply arrivals by 2, 4, or 8. Reward is normalised by each step's pending-request count, and Fig. S4 plots the reward each policy keeps.

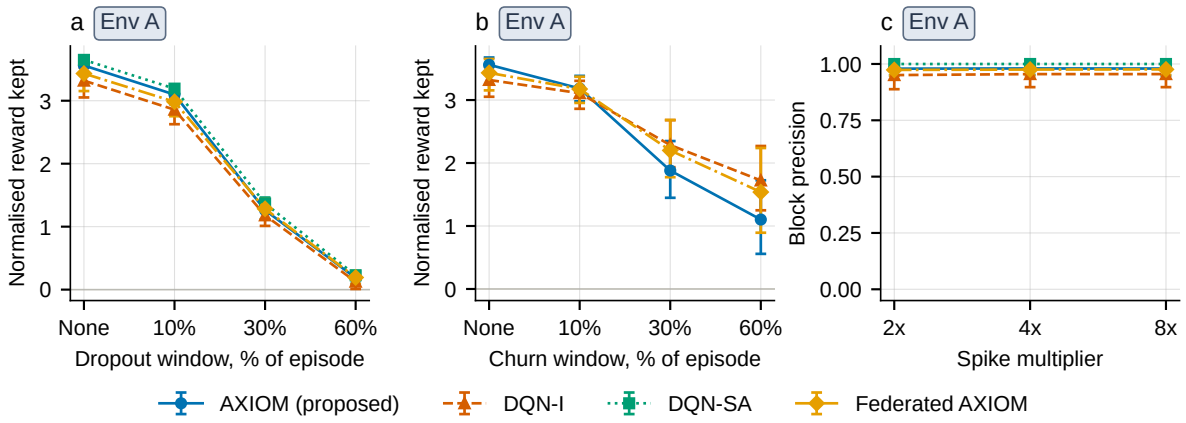


Fig. S4 Disruption campaign, 10 seeds per policy, 95% CIs. a, b, Normalised reward kept under dropout and churn, with the undisrupted level at the tick marked None. c, Block precision under demand spikes

Dropout and churn hurt every policy significantly at every severity, in 9 or 10 of 10 seeds at Holm-adjusted $p \leq 0.041$, and no gap between AXIOM and any baseline survives correction. Demand spikes leave AXIOM's decisions largely intact, with block precision at 0.98, against 1.00 for DQN-SA and about 0.95 for DQN-I.

S6. Scaling to thirty-seven gNBs and stronger baselines

In the gate map of the main paper's thirty-seven-gNB cluster, DQN-SA leaves every admission gate open at 0.93 to 0.95 and DQN-I closes the mMTC gate only to 0.94. MAPPO there partly closes the eMBB gate to 0.80 but leaves mMTC open at 0.51. At seven gNBs AXIOM keeps URLLC and eMBB at 99.8% and 94.4%, against 91.4% and 61.0% for DQN-SA, 94.2% and 55.0% for DQN-I, and 87.0% and 48.4% for MAPPO. At nineteen gNBs it keeps them at 99.4% and 95.1%, and at thirty-seven at 96.7% and 90.7%.

The thirty-seven-gNB configuration extends the nineteen-gNB one exactly as the seven- and nineteen-gNB configurations extended three, with 37 arrivals per step and at most 148 pending requests. AXIOM ran 30 seeds on the hex layout and 15 on each of the ring and fully-connected layouts. In Env A the baselines and the static rule never read the layout, so each ran once on the 30 hex seeds, and Table S2 lists every policy.

AXIOM's six collapsed runs keep 59% to 65% priority-slice compliance, while its other runs keep 96.9% to 98.1% on average under each layout. We also ran MAPPO with the configuration that the main paper's grid selected for Env A, trained for 900 episodes. It keeps 65.6% priority-slice compliance and 151.90 reward per step and never collapses. AXIOM beats it by 29.2 points in 13 of 15 seeds at $p < 0.001$ and by 23.23 reward per step in all 15.

The fresh-seed replication retrained every policy on seeds 6100 to 6119 on the hex layout, as the lower part of Table S2 lists. AXIOM leads DQN-SA by 37.7 points of priority-slice compliance and 35.46 reward per step, DQN-I by 61.0 and 25.62, and MAPPO by 36.5 and 30.63. It wins 18 to 19 of 20 seeds in every comparison, all at Holm-adjusted $p < 0.001$. Its one collapsed seed is 6110, and pooled with the 30 hex seeds above, AXIOM keeps 94.2% priority-slice compliance over 50 seeds and collapses in 4 of them.

Accepting every request is the floor that any admission controller should clear. In Env A it earns 26.88, 78.55, and 156.22 reward per step at seven, nineteen, and thirty-seven gNBs. At nineteen and thirty-seven gNBs every baseline keeps less priority-slice compliance than accept-all on average, and DQN-SA's runs that do not collapse keep only 40.0% and 35.9%, DQN-I's 55.9% and 29.1%.

Table S2 Env A with thirty-seven gNBs, reported and fresh seeds. Priority-slice compliance is the mean of URLLC and eMBB compliance, and collapse counts policies that accept every request

Policy	Runs	URLLC	eMBB	Priority SLA	Reward	Collapsed
AXIOM, hex	30	97.4%	89.8%	93.6%	174.20	3
AXIOM, ring	15	93.7%	92.6%	93.2%	173.51	2
AXIOM, fully-connected	15	98.4%	90.4%	94.4%	174.44	1
DQN-SA	30	73.3%	44.1%	58.7%	144.70	26
DQN-I	30	38.6%	25.7%	32.1%	148.35	3
MAPPO	30	67.8%	45.6%	56.7%	145.25	0
Static rule	30	100.0%	100.0%	99.98%	177.58	0
Fresh seeds 6100 to 6119, hex layout						
AXIOM	20	95.4%	94.7%	95.1%	174.67	1
DQN-SA	20	71.6%	43.1%	57.4%	139.20	18
DQN-I	20	42.0%	26.1%	34.1%	149.04	2
MAPPO	20	70.1%	47.0%	58.5%	144.04	0
Static rule	20	100.0%	100.0%	99.97%	177.68	0

The DQN baselines were tuned on pilot seeds over learning rates from 10^{-4} to 10^{-3} and minibatches of 16 and 64, and retrained on the 30 hex seeds wherever a new setting won, as Table S3 lists. The grid selected new settings in three of the four combinations of baseline and size. No tuned baseline exceeds the accept-all floor at either size. At nineteen gNBs the tuned DQN-SA accepts every request in all 30 evaluation seeds. AXIOM beats each tuned baseline on compliance and reward in every setting, winning 26 to 30 of 30 seeds at Holm-adjusted $p < 0.001$.

Table S3 DQN baselines tuned on pilot seeds, Env A, 30 hex seeds. LR is the learning rate, batch the minibatch size, and Before the priority-slice compliance before tuning

Baseline	gNBs	LR, batch	Priority SLA	Before	Reward	Collapsed
DQN-SA	19	10^{-4}, 16	67.1%	65.3%	78.55	30
DQN-I	19	$3{\times}10^{-4}$, 64	59.1%	56.4%	77.07	11
DQN-SA	37	10^{-3}, 16, unchanged	58.7%	58.7%	144.70	26
DQN-I	37	10^{-4}, 64	45.5%	32.1%	150.42	18

The value-decomposition network (VDN) trains DQN-I's per-gNB networks on one joint target per step, with no tuning. VDN ties AXIOM at three gNBs in both environments but falls behind at nineteen gNBs, as Table S4 shows. There AXIOM keeps 13.6 percentage points more and earns 2.73 more reward per step, winning 27 to 28 of 30 seeds at Holm-adjusted $p < 0.001$. At 110% and 120% of the nominal load at nineteen gNBs, AXIOM keeps 24.3 to 34.0 and 2.5 to 6.4 percentage points more priority-slice compliance than every learned baseline, at Holm-adjusted $p \leq 0.006$. At 120% the priority demand exceeds the slice caps at most gNBs, so no policy can protect it throughout.

Table S4 VDN against AXIOM and DQN-I, reward per step and priority-slice compliance, 30 seeds per cell

Cell	VDN	AXIOM	DQN-I	VDN SLA	AXIOM SLA	DQN-I SLA
Env A, 3 gNBs	12.664	12.698	11.587	99.0%	98.9%	92.9%
Env B, 3 gNBs	8.552	8.832	7.794	62.5%	65.1%	59.0%
Env A, 19 gNBs, hex	86.428	89.161	76.388	83.4%	97.0%	56.4%

Fig. S5 gives the per-layout view of the Env A scaling results that the main paper pools over layouts.

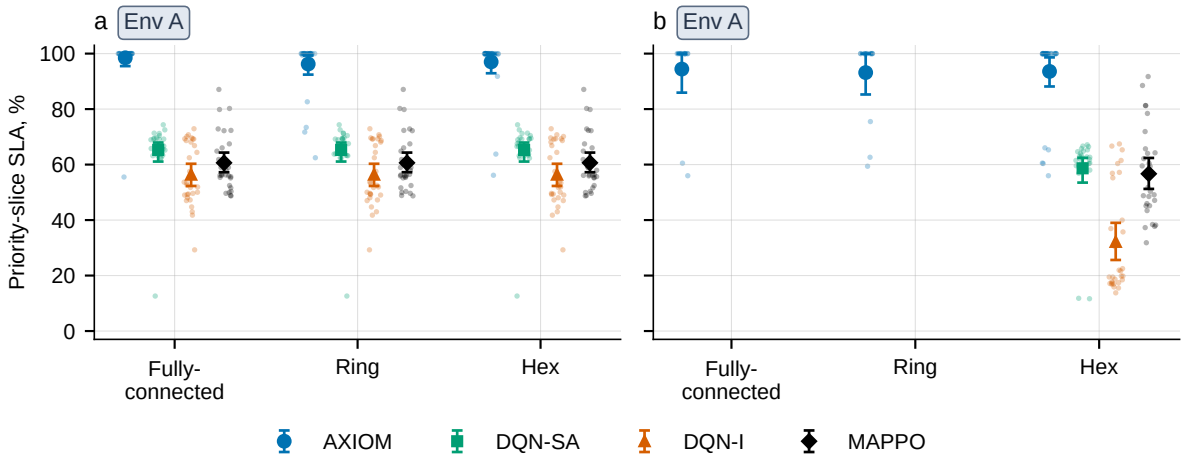


Fig. S5 Env A priority-slice compliance by layout, per-seed points behind the mean and 95% CI. a, Nineteen gNBs, 30 seeds per layout. b, Thirty-seven gNBs, where the baselines never read the layout and ran once, on the hex seeds

S7. Env B, shifted load, longer training, and convergence

No deployable policy we tested keeps eMBB compliance above 37% in Env B, because demand at the load peaks exceeds the budget by up to 23.5%. Even the rule with the highest eMBB compliance among all 95 local rules, chosen in hindsight on each evaluation stream, reaches only 37.7% on average. The best rule by reward in hindsight earns only 0.14% more than the tuned rule.

Table S5 lists AXIOM against the rule tuned on nominal conditions, on 25 checkpoints neither re-trained nor re-tuned. A rule re-tuned on the shifted training streams selected thresholds different from the nominal rule's at every shift. Against it AXIOM trails by 0.16 and 0.12 per step at 110% and 115% load, which is not significant, and by 0.62 at 120% load. Under 110% to 120% load AXIOM beats the nominally tuned rule in 21 to 23 of 25 seeds at Holm-adjusted $p \leq 0.0002$. At seven gNBs on the ring layout the reversal is larger. AXIOM trails the rule by 3.84 reward per step under nominal load but leads it by 1.86 and 3.14 at 110% and 120% load.

Table S5 AXIOM against the rule tuned on nominal conditions, Env B with three gNBs, 25 seeds, reward per step, wins over losses, and the unadjusted Wilcoxon p

Condition	AXIOM	Tuned rule	Difference	AXIOM wins	p
80% load	13.776	14.055	−0.279	12/13	0.44
nominal	8.846	9.261	−0.415	7/18	0.00016
105% load	4.482	4.183	+0.299	21/4	0.017
110% load	2.082	1.624	+0.458	23/2	8.8e-05
115% load	0.378	−0.052	+0.431	23/2	1e-05
120% load	−0.714	−1.068	+0.354	21/4	8.8e-05
coupling 0.30	6.457	6.766	−0.310	11/14	0.28
load swing 0.50	6.015	5.831	+0.184	18/7	0.045

With three times the training budget AXIOM gains 0.33 reward per step at three gNBs and 2.92 at seven, on 20 and 12 seeds. The remaining gaps to the tuned rule, 0.08 and 0.96 per step, are no longer significant, at $p = 0.35$ and $p = 0.064$.

Table S6 lists Env B at nineteen gNBs on the hex layout, over 15 seeds. Per-gNB compliance counts a slice as compliant at a gNB and step when that gNB meets the slice's SLA, whereas the cluster-level metric counts a violation at any gNB as a violation of the cluster. AXIOM makes exactly the static rule's decisions in 9 of the 15 seeds, and in 5 of the other 6 it also blocks only mMTC.

Further runs answer an internal review, and four of their six predictions failed. The per-gNB penalty charges each slice's violation penalty once for every violating gNB, in training and evaluation. Under uneven load each gNB's load multiplier is drawn between 75% and 125% per seed, and the rule re-tuned on those streams sheds eMBB once the URLLC queue passes 0.05 rather than 0.2. Under this uneven

Table S6 Env B with nineteen gNBs, 15 seeds, hex layout. SLA columns give priority-slice compliance, the mean of URLLC and eMBB compliance, per gNB and at cluster level

Policy	Reward	Per-gNB SLA	Cluster SLA	Collapsed
AXIOM	80.28	77.3%	24.7%	1
DQN-SA	74.74	66.4%	18.0%	14
DQN-I	71.20	66.8%	16.7%	0
MAPPO	59.07	71.8%	20.8%	0
Static rule	81.59	79.9%	23.0%	0
Accept all	74.77	66.4%	18.0%	15
Tuned local rule, carried over	84.09	84.5%	55.7%	0

load at seven gNBs AXIOM does not beat the nominally tuned rule, earning 0.36 more reward per step at $p = 0.23$. It still beats every learned baseline there, by 1.18 to 4.28 reward per step at unadjusted $p \leq 0.005$. Against MAPPO under uneven per-gNB load at seven gNBs it earns 27% more and wins every seed. Charging the SLA penalty per violating gNB, so that it grows with the cluster, raised AXIOM's per-gNB compliance by only 1.9 points, at $p = 0.65$. Under that reward every learned baseline kept 13.7 to 18.5 percentage points less per-gNB compliance than AXIOM, in every seed.

Two further runs explain AXIOM's position at seven gNBs, and every one of their checks held. At seven gNBs the tuned rules earn 26.5 to 27.3 per step and beat AXIOM in all 15 seeds under every layout, while AXIOM earns 23.4 to 23.6, no more than the static rule's 23.6. At seven gNBs 28 of AXIOM's 45 seeds shed only mMTC, which is the static rule's policy. None of AXIOM's 30 seeds does this at three gNBs. The reward charges each slice's SLA penalty once for the cluster when any gNB violates it, so shedding eMBB at one gNB pays only when the other gNBs also comply.

Table S7 lists three changes, each tested on the ring layout with 15 seeds. Compared with the original setting, the per-gNB SLA penalty raised AXIOM's compliance by 10.7 points, a learning rate of 0.003 by 14.9, and 900 training episodes by 21.7. The Holm-adjusted p values are 0.007, 0.007, and 0.0004. The per-gNB penalty changes the reward itself, so only compliance compares there. Under that reward AXIOM keeps 20.9 points more than DQN-SA trained the same way, which collapses in 11 of 15 seeds.

Table S7 Env B, seven gNBs, ring layout, 15 seeds, reward per gNB per step, priority-slice compliance, and seeds that shed only mMTC

Policy	Reward per gNB	Priority SLA	mMTC-only seeds
AXIOM, 300 episodes, as reported	3.36	34.0%	11
AXIOM, per-gNB SLA penalty	–	44.7%	0
AXIOM, learning rate 0.003	3.50	48.9%	3
AXIOM, 900 episodes	3.79	55.7%	1
DQN-SA, 900 episodes	3.35	41.9%	1
DQN-I, 900 episodes	3.29	42.9%	0
MAPPO, 900 episodes	2.97	40.5%	0
Tuned local rule	3.91	60.5%	0

With 900 episodes for every learner, AXIOM keeps 12.8 to 15.2 points more compliance and earns 0.44 to 0.82 more reward per gNB than each baseline, at Holm-adjusted $p \leq 0.0012$. The baselines gain at most 7.0 points from the longer budget. The tuned rule still keeps more compliance than AXIOM in all 15 seeds. At nineteen gNBs the longer budget does not help. After 900 episodes seven of eight seeds still shed only mMTC, against all eight after 300.

Fig. S6 shows what AXIOM learned at seven gNBs on the ring layout. After 300 episodes the eMBB head of 11 of its 15 seeds would reject at no step, and pooled over all 15 it rejects on 2.0% of gNB-steps. After 900 episodes 14 of 15 seeds reject more often as the gNB's own URLLC queue grows. Pooled, the share rises from 12.2% below a queue of 0.05 to 71.8% between 0.75 and 1.0, a softer form of the tuned rule's step at 0.2.

On 15 fresh seeds, 7000 to 7014, in Env B at nineteen gNBs the results above held again. AXIOM earned 5.90 to 22.00 more reward per step than each learned baseline and kept 6.9 to 11.1 points more per-gNB priority-slice compliance, at Holm-adjusted $p \leq 0.011$, keeping 76.9%. The static rule again earned more, by 0.87 per step, and tied AXIOM exactly in 13 of the 15 seeds. The local rule carried over from seven gNBs won every seed, by 3.33 per step, and DQN-SA collapsed in all 15.

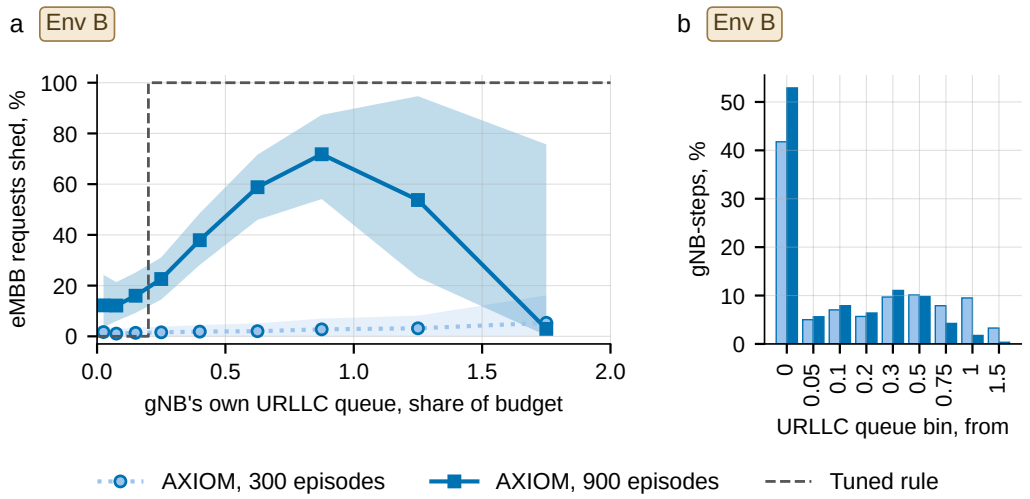


Fig. S6 What AXIOM learned in Env B at seven gNBs on the ring layout, 15 seeds. a, Share of gNB-steps on which its eMBB head would reject, by the gNB's own URLLC queue, with 95% CIs over seeds. b, How often each queue level occurs

S8. Throughput and delay

In Env B at nineteen gNBs AXIOM's URLLC delay is below every learned baseline's but above the tuned rule's. Table S8 lists throughput, the share of offered demand that is served, and delay, the mean backlog over the mean served PRBs in 5 s control periods, with the 95th percentile over steps and gNBs. This delay is a queueing proxy from the fluid model, not packet latency. Every stored checkpoint and rule was re-evaluated with a recorder under a plan committed beforehand. Each re-run with a stored log reproduced it exactly, and the Env A accept-all rows, which have no stored log, are marked unchecked.

Table S8 Served share of offered demand and backlog delay in control periods, mean and 95th percentile, per slice

Policy	URLLC served	eMBB served	URLLC delay	p95	eMBB delay	p95
Env A, three gNBs						
AXIOM	99.8%	99.7%	0.11	0.66	0.11	0.47
DQN-SA	99.5%	99.4%	0.20	0.96	0.18	0.83
DQN-I	99.4%	98.5%	0.23	1.11	0.43	6.24
MAPPO	98.4%	90.5%	0.38	3.31	3.81	19.67
Static rule	99.8%	99.9%	0.10	0.59	0.05	0.31
Accept all, unchecked	97.5%	97.6%	0.80	3.01	0.76	2.65
Env A, thirty-seven gNBs, hex						
AXIOM	99.8%	99.0%	0.08	0.42	0.34	1.12
DQN-SA	94.2%	95.2%	3.61	16.87	2.09	14.28
DQN-I	97.7%	97.5%	0.76	5.23	0.72	5.49
MAPPO	96.4%	85.7%	0.58	5.38	6.28	35.70
Static rule	100.0%	100.0%	0.03	0.17	0.02	0.09
Accept all, unchecked	98.7%	98.8%	0.43	2.14	0.38	1.83
Env B, three gNBs						
AXIOM	98.4%	88.5%	0.65	2.32	3.18	19.28
DQN-SA	98.2%	82.1%	0.75	3.15	5.73	49.79
DQN-I	98.6%	82.7%	0.66	2.81	4.73	49.34
MAPPO	97.2%	83.5%	1.10	4.00	4.57	43.82
Static rule	96.0%	96.1%	1.59	5.25	1.57	5.11
Tuned rule	99.0%	84.8%	0.58	1.96	3.57	38.55
Accept all	92.8%	92.9%	2.59	7.72	2.58	7.59
Env B, nineteen gNBs, hex						
AXIOM	96.6%	92.6%	1.35	4.67	3.41	16.01
DQN-SA	93.8%	93.9%	2.26	6.99	2.24	6.82
DQN-I	94.1%	86.1%	2.17	9.71	4.17	72.77
MAPPO	89.2%	88.3%	2.66	33.84	2.58	28.13
Static rule	96.8%	96.9%	1.33	4.65	1.32	4.50
Tuned rule, carried	99.1%	87.9%	0.51	1.77	2.78	25.40
Accept all	93.8%	93.9%	2.25	6.97	2.24	6.82

S9. Reproducing the results

Fig. S7 sets out what reproducing the results needs and the order of the steps. Every result in both documents comes from one laptop without a graphics processor, and every simulation run fixes its seed. Reruns on the same seeds reproduce the stored per-seed results exactly, except the original DQN-SA runs, whose initial weights were not seeded, as Section S2 explains. The code, configurations, and stored logs are available from the corresponding author on reasonable request.

What is needed

Hardware
One laptop, Intel Core i5-1135G7, 4 cores and 8 threads
8 GB of memory, 7.4 GiB usable, with 23 GiB of swap
No graphics processor, every run on the CPU
Ubuntu 24.04 LTS

Software
Python 3.12.3, PyTorch 2.13.0 on the CPU
NumPy 2.5.1, SciPy 1.18.0, Matplotlib 3.11.0, PyYAML 6.0.3
TeX Live 2023 with pdfLaTeX
Simulators and learners in `framework/`, never modified

Live testbed check only
OAI gNB, ORANSlice fork on OAI 2024.w28, rfsim radio, band n78, 106 PRBs
Open5GS core in Docker, 17 containers
protobuf-c 1.5.2, built from source

Steps

1. **Set up.** Request the code, configurations, and stored logs from the corresponding author. `framework/` holds the simulators and learners, `experiments/scripts/` the campaign scripts, `experiments/configs/` the cluster configurations, `experiments/results/` the stored per-seed logs, and `docs/` the pre-registrations. Create a Python virtual environment with the packages above. Every job runs from `framework/` on one thread, with `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, and `OPENBLAS_NUM_THREADS=1`.

2. **Train and evaluate.** One command trains and evaluates one policy on a batch of seeds, for example
`M63_BACKLOG_RESET=1 M64_AXIOM_V2=1 python m64_run.py --arm axiom2 --env A --n 19 --topology ring --seeds 900 901 902 903 904 --out-dir ../experiments/results/m65_scaling/n19_ring`
The arms are `axiom2`, `single_agent_dqn`, `independent_dqn`, `mappo_tuned`, and the rules, whose thresholds are chosen on training streams. Each seed trains for 300 episodes on its own stream and is then evaluated for 50 episodes on the held-out stream 5000 plus the seed. Each seed writes a training log, a checkpoint, and an evaluation log, and each batch a JSON file of reward, blocks, and training time. The median training time per seed runs from 73 s to 2,088 s, and the main campaigns ran five jobs at a time. The primary seeds are 900 to 929, the replication 1000 to 1009, the pilots 2000 to 2009, and the fresh seeds 3000 onwards.

3. **Run the other campaigns.** `m3_run_experiment.py`, called through `m62_shared_env.py`, trains the federated variant with noise multipliers 0, 0.5, 1, 2, and 4. `m4_run_experiment.py` evaluates frozen checkpoints under dropout, spikes, and churn. `m73_run.py` adds the 37-gNB configuration, and `m70_run.py`, `m75_run.py`, and `m76_run.py` add the follow-up variants, selected by environment variables. `m63_validation_screen.py` screens every trained checkpoint on its own training stream.

4. **Re-evaluate stored checkpoints.** `m72_shift.py eval` replays AXIOM and the nominally tuned rule at 80% to 120% load, stronger coupling, and a wider load swing, and `m72_shift.py retune` re-tunes the rule on shifted training streams. `m77_kpi.py` records the PRBs served and the backlog, and confirms that each re-run reproduces its stored log. `m81_decision_probe.py` replays every stored evaluation to read what the eMBB head decides.

5. **Summarise.** `m65_rebuild_all.sh` runs `m64_analysis.py` and `m62_analysis.py` over the main campaigns, and each later campaign has its own `analysis` script. Each writes the summaries and judges every pre-registered criterion in `docs/PAPER5_M*_preregistration.md`, reporting each whether it holds or fails.

6. **Verify.** `m66_verify.py --tex main.tex --supp supplementary.tex` reads the raw logs with its own reader and Holm correction, and recomputes every results table and key sentence of both documents. Its `m82_audit.py` traces every remaining number to the configurations and the code. `m66_ci_recompute.py` recomputes every bootstrap CI with 10,000 resamples, and `m66_v1v2.py` reruns reported settings in fresh processes.

7. **Redraw and build.** `paper5_regen_wpc_figures.sh` redraws every plotted figure from the stored results, with `MATPLOTLIBRC=experiments/plots`. `paper5_cvd_check.py` renders each figure in greyscale and as seen with protan, deutan, and tritan colour vision. `make_submission.py` writes `Fig1` to `Fig8`, `ESM_1.pdf`, and the compiled manuscript.

Live testbed

Live testbed check. Start the Open5GS core with Docker Compose from `docker_open5gs/5g-sa-deploy-slicing.yaml`. Start the gNB with
`nr-softmodem -O ORANSlice.gnb.sa.band78.fr1.106PRB.usrpx310.conf --sa --rfsim`
and up to six `nr-uesoftmodem` UEs, from the second on each in its own network namespace. Drive iperf3 UDP traffic on every slice. `run_live_eval_arm.py` then runs the checkpoint through the UDP E2 agent on ports 6600 and 6655, in short batches with a health check, and a stack restart if needed, before each batch.

Fig. S7 Reproducing the results. Top, the hardware and software every run used. Below, the steps from the stored code to the figures and the manuscript, and the live testbed check